ESAM 420 Notes

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1 Motivating Examples

Suppose we want to launch a projectile vertically into the air and we want to create a model. We are launching it vertically so we do not have to worry about vectors. We start with Newton's Law of Gravity so we can say that

$$\frac{d^2r}{dt^2} = -G\frac{M_e}{r^2}$$

where r is the distance from the center of the Earth, G is the gravitational constant, and M_e is the mass of the Earth. However, we want to know how far from the surface of the Earth our projectile goes so we define $r = h + R$ where R is the radius of the Earth and h is the height of the projectile relative to the ground. Since $dr = dh$, we have that

$$\frac{d^2h}{dt^2} = -G\frac{M_e}{(h+R)^2}$$

Recalling that $g = \frac{GM_e}{R^2}$, we rewrite the equation as

$$\frac{d^2h}{dt^2} = -\frac{gR^2}{(h+R)^2}$$

This equation has no analytical solution. However, if make the approximation $h \ll R$, we can obtain an *asymptotic solution.*

$$\frac{d^2h}{dt^2} = -g$$

which if we set the initial conditions to be $h(0) = 0, h'(0) = v_0$, we have that the solution is

$$h(t) = v_0 t - \frac{1}{2}gt^2$$

The problem with this is we have no semi-"precise" way to justify when we can make this assumption. We also cannot add any correcting factors. We usually start these by non-dimensionalizing. This allows us to zoom to the scale we want to work with and justify what is "small." For the gravity problem, let us define the characteristic length and times to be

$$L_c = \frac{v_0^2}{g} \qquad T_c = \frac{v_0}{g}$$

We also define some new normalized variables relative to these characteristic norms (I don't want to use variables and they're mainly about magnitudes, hence norms)

$$x = \frac{h}{L_c} \qquad \tau = \frac{t}{T_c}$$

We can transform $\frac{d^2h}{dt^2} \to \frac{d^2x}{d\tau^2}$ via the chain rule

$$\begin{aligned}
\frac{dx}{d\tau} &= \frac{dx}{dh}\frac{dh}{dt}\frac{dt}{d\tau} = \frac{1}{L_c}\frac{dh}{dt}\frac{T_c}{1} \\
\frac{dh}{dt} &= \frac{L_c}{T_c}\frac{dx}{d\tau} \\
\Rightarrow \frac{d^2h}{dt^2} &= \frac{d}{dt}\left(\frac{L_c}{T_c}\frac{dx}{d\tau}\right) = \frac{d}{dt}\frac{dt}{d\tau}\left(\frac{L_c}{T_c}\frac{dx}{d\tau}\right) = \frac{L_c}{T_c^2}\frac{d^2x}{d\tau^2}
\end{aligned}$$

So our equation now becomes

$$\begin{aligned}
\frac{L_c}{T_c^2}\frac{d^2x}{d\tau^2} &= -\frac{gR^2}{(xL_c + R)^2} = -\frac{gR^2}{R^2(x\frac{L_c}{R} + 1)^2} \\
\frac{d^2x}{d\tau^2} &= -\frac{1}{(x\varepsilon + 1)^2} \qquad \varepsilon = \frac{L_c}{R} = \frac{v_0^2}{gR}
\end{aligned}$$

We then guess our solution is some sort of expansion based off this small parameter of the form

$$x(t) = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

Our small parameter being the ratio of the characteristic length to the radius of the Earth shows us exactly when we can make certain assumptions.

Another application is the Navier-Stokes equations.

$$\nabla \cdot \vec{u} = 0 \text{ (incompressibility equation)}$$

$$\rho\left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\,\vec{u}\right) = -\nabla p + \mu\nabla^2 u \text{ (equation of motion)}$$

If we non-sensationalize such that $(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{L}(x, y, z), \tilde{\vec{u}} = \frac{\vec{u}}{u_c}, \tilde{p} = \frac{p}{p_c}, \tilde{t} = \frac{t}{T_c}, T_c = \frac{L_c}{u_c}$. This transforms our equation into

$$\frac{\rho u_c}{T_c}\frac{\partial \tilde{\vec{u}}}{\partial \tilde{t}} + \frac{\rho u_c^2}{L_c}\left(\tilde{\vec{u}} \cdot \tilde{\nabla}\right)\tilde{\vec{u}} = -\frac{p_c}{L_c}\tilde{\nabla}\tilde{p} + \frac{\mu U_c}{L_c^2}\tilde{\nabla}^2\tilde{u}$$

Notice that the first two coefficients of the left hand side are equal and so we can rewrite the left hand side as one differential operator $\frac{D\vec{u}}{D\tilde{t}}$. Doing some algebra we get that

$$\mathrm{Re}\,\frac{D\vec{u}}{Dt} = -\frac{p_c L_c}{\mu u_c}\tilde{\nabla}p + \tilde{\nabla}^2\tilde{u}$$

where Re is the Reynold's number which is the ratio of inertial versus viscous forces in a flow.

$$\mathrm{Re} = \frac{\rho u_c L_c}{\mu}$$

If we take the limit as the Reynold's number go to zero, we go into the length scale of bacteria typically. If we have $\frac{p_c L_c}{\mu u_c} = 1$, we then have that the characteristic pressure is $p_c = \frac{\mu u_c}{L_c}$. Rearranging the equation we get

$$\nabla p = \mu \nabla^2 \vec{u}$$

which is known as the Stoke's limit. If instead we take the limit as the Reynold's number goes to infinity and if we set the characteristic pressure to be $p_c = \rho u_c^2$, we get

$$\frac{D\vec{u}}{D\tilde{t}} = -\nabla p + \frac{1}{\mathrm{Re}} \tilde{\nabla}^2 \tilde{u}$$

The Laplacian term goes to zero and we have the Euler Equation for invsicid flow

$$\rho \left(\frac{\partial u}{\partial t} + (\vec{u} \cdot \nabla)\vec{u} \right) = -\nabla p$$

From this we can get d'Alembert's paradox which states that the force of drag is zero on a body moving at a constant velocity in an inviscid fluid contrary to observations. Due to asymptotics, we can add correction terms to resolve this paradox.

2 Algebraic Equations

Suppose we want to solve the algebraic equation

$$x^2 + 2\varepsilon x - 1 = 0 \qquad \varepsilon \ll 1$$

Analytically we can solve this to get that $x_{1,2} = -\varepsilon \pm \sqrt{1+\varepsilon^2}$. If we can't analytically solve, we can use two methods to approximate: expansion or iteration. Let us start with the expansion method first.

2.1 Expansion Method

Let us start with the zeroth-order approximation and set $\varepsilon = 0$. We get that our solutions are $x_{0_{1,2}} = \pm 1$. We then make the guess that our solution can be expressed as the power series

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \ldots$$

We then plug that back into our equation and match together powers of ε up until the powers of epsilon that we desire. Doing so gets us

$$\begin{aligned} (x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \ldots)^2 + 2\varepsilon(x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \ldots) - 1 = 0 \\ 1 + 2\varepsilon x_1 + 2\varepsilon^2 x_2 + \varepsilon^2 x_1 + \ldots + 2\varepsilon + 2\varepsilon^2 x_1 + \ldots - 1 = 0 \end{aligned}$$

Grouping them together by powers of epsilon gives

$$\varepsilon(2x_1 + 2) + \varepsilon^2(2x_2 + x_1^2 + 2x_1) = 0 \ \ \forall \varepsilon$$

Since this equation must be true for all ε, we have that the parenthesis terms must be 0 and so we solve for x_1 and then plug that into the ε^2 expression to solve for x_2.

$$\begin{aligned} \varepsilon: \ 2x_1 + 2 = 0 \rightarrow x_1 = -1 \\ \varepsilon^2: \ 2x_2 + 1 - 2 = 0 \rightarrow x_2 = \frac{1}{2} \end{aligned}$$

Our three-term expansion is therefore

$$x = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

If we take the binomial expansion of our analytical solution, we find the same thing.

2.2 Iteration Method

We generally solve for the highest order of x

$$x^2 = 1 - 2\varepsilon x \to x = \pm\sqrt{1 - 2\varepsilon x}$$

The iteration method says that

$$x_{n+1} = \pm\sqrt{1 - 2\varepsilon x_n}$$

We start with the zeroth-order approximation

$$\begin{aligned} \varepsilon = 0 \to x_0 &= \pm 1 \\ \varepsilon^1 \quad x_1 &= \sqrt{1 - 2\varepsilon} = 1 - \varepsilon \\ \varepsilon^2 \quad x_2 &= \sqrt{1 - 2(1-\varepsilon)\varepsilon} = 1 - \varepsilon(1-\varepsilon) - \frac{1}{8}(2\varepsilon(1-\varepsilon))^2 \end{aligned}$$

Cleaning all of this up we get the same thing as the previous methods

$$x_3 = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

The iteration method effectively is a spiral from an initial approximation leading to a value of x_n which we then iterate on more to hopefully get to the desired root. However, if the function changes too much too quickly, this can lead to spiraling away.

The expansion and iteration methods are good when the zeroth-order approximation doesn't change the nature of the equation (e.g., if we have $\varepsilon x^2 + x + 1 = 0$, the zeroth-order approximation would change the equation from a quadratic to a linear equation).

Let us now consider a case when we can't use the highest order term to begin our iteration.

$$x^2 + x - \varepsilon = 0$$

The key thing here is to setup and iteration such that it converges to the desired value. In order for that to happen, we need on an interval I around the root to have the following condition: $|f'(x)| < 1 \ \ \forall x \in I$. It is also worth noting that knowing what the unperturbed function looks like by moving all of the ε terms to one side and ignoring it (in this case, it would be $x^2 + x = 0$). Here we can see the unperturbed roots are $x = -1$ and $x = 0$. Let us try to find the root around $x = -1$. We can do this with the setup

$$x^2 = \varepsilon - x$$

By dividing by x, we can find the relation

$$x = \frac{\varepsilon}{x_n} - 1$$

If we let the right hand side equal $f(x)$, we find that the derivative is $|f'(x)| = \frac{\varepsilon}{x_n^2}$. Since ε is assumed small, our convergence condition is satisfied.

2.3 Method of Dominant Balance

Let us take a look at the algebraic equation

$$\varepsilon x^2 + x - 1 = 0$$

If we do either the expansion or iteration method, we change the number of roots we get. Let us go try out this method to see what happens. We once again start with the $\varepsilon = 0$ term to get that $a_0 = 1$. We shall use expansion here and guess $x = 1 + \varepsilon a_1 + \varepsilon^2 a_2 + \ldots$. Plugging this in we get

$$\begin{aligned} \varepsilon(1 + \varepsilon a_1 + \ldots)^2 + (1 + \varepsilon a_1 + \ldots) - 1 &= 0 \\ \varepsilon(1 + 2\varepsilon a_1 + \ldots) + (1 + \varepsilon a_1 + \ldots) - 1 &= 0 \\ \varepsilon(1 + a_1) + \mathcal{O}(\varepsilon^2) = 0 \to a_1 &= -1 \end{aligned}$$

Therefore our two term expansion is $x = 1 - \varepsilon + \mathcal{O}(\varepsilon^2)$. If we look at the actual roots using the quadratic formula and binomial expand the square root, we get the same thing for the positive root.

$$x = \frac{-1 \pm \sqrt{1+4\varepsilon}}{2\varepsilon} = \frac{1}{2\varepsilon}\left(-1 \pm (1 + \frac{1}{2}4\varepsilon - \frac{1}{8}(4\varepsilon)^2 + \dots\right)$$
$$x_+ = \frac{1}{2\varepsilon}(2\varepsilon - 2\varepsilon^2) = 1 - \varepsilon + \dots$$

If we look at the negative root, we find that there is a $\frac{1}{\varepsilon}$ term which explains the behavior at $\varepsilon = 0$ and why it doesn't show up.

$$x_- = \frac{1}{2\varepsilon}(-2 - 2\varepsilon + 2\varepsilon^2) = -\frac{1}{\varepsilon} - 1 + \varepsilon + \dots$$

equipped with this knowledge, let us introduce the Method of Dominant Balance. This method allows us to rigorously find the other root by introducing a rescaled variable $y = \frac{x}{\delta}$ where δ is some unknown which may be a function. This relation implies that $x = \delta y$. The reason we are doing this is because in the original equation we are strictly assuming $x = \text{ord}(1)$ with respect to ε and so as such, the linear and constant terms are both order 1 whereas the quadratic term is very small. As such, the dominant balance was between x and -1 and that's why we only recovered one of the roots. This rescaling shifts the dominant balance. Since we know before hand that the irregular root $x \sim \frac{1}{\varepsilon}$, let us plug into the original equation and see what happens.

$$\varepsilon\left(\frac{1}{\varepsilon}\right)^2 + \left(\frac{1}{\varepsilon}\right) - 1 = 0$$
$$\frac{1}{\varepsilon} + \frac{1}{\varepsilon} - 1 = 0$$

We can now see that as $\varepsilon \to 0$, the quadratic and linear terms are now big compared to the constant. That is effectively what the Method of Dominant Balance is doing except we do not have prior knowledge of what this relation is. As such, we substitute our relation for x to arrive at

$$\varepsilon\delta^2 y^2 + \delta y - 1 = 0$$

and now we balance the two other terms (with the assumption that y is strictly order 1). For this to happen, we must have the relation

$$\varepsilon\delta^2 \sim \delta$$

which we can then extract the relation $\delta \sim \frac{1}{\varepsilon}$. This is exactly the term that caused the negative root to vanish when we used the quadratic formula to solve for it. We now have the relation $x = \frac{1}{\varepsilon}y$. We are now able to substitute this back in for every x in the original algebraic equation and derive a new one that we can apply our previous methods on.

$$\frac{1}{\varepsilon}y^2 + \frac{1}{\varepsilon}y - 1 = 0$$
$$\Rightarrow y^2 + y - \varepsilon = 0$$

We cannot find the solutions to this using iteration because we would have the relation $y = \pm\sqrt{\varepsilon - y}$ which if we tried to find the leading term expression, we would result in complex values. That is why we shall use expansion

$$y = a_0 + \varepsilon a_1 + \varepsilon^2 a_2 + \dots$$
$$(a_0 + \varepsilon a_1 + \dots)^2 + (a_0 + \varepsilon a_1 + \dots) - \varepsilon = 0$$
$$a_0^2 + 2\varepsilon a_0 a_1 + a_0 + \varepsilon a_1 - \varepsilon = 0$$
$$a_0(a_0 + 1) + \varepsilon(2a_0 a_1 + a_1 - 1) = 0$$

Solving for a_1 in the ε term we see that

$$a_1 = \frac{1}{2a_0 + 1}$$

We see that $a_0 = 0$ or $a_0 = -1$. In the first case, we have that $a_1 = 1$ and so the expansion for that is $y = \varepsilon + \dots$ and for the second, we have $-1 - \varepsilon + \dots$. Subbing this back into the relation for $x = \frac{1}{\varepsilon}y$, we get the approximations

$$x_1 = 1 + \dots$$
$$x_2 = -\frac{1}{\varepsilon} - 1 + \dots$$

Let us look at another example

$$\varepsilon x^3 - x + 1 = 0$$

Finding the regular root is pretty trivial so we shall ignore it. We start with the ansatz that $x = \delta y$. Subbing this in we get

$$\varepsilon \delta^3 y^3 - \delta y + 1 = 0$$

We can choose to either balance the first and second term or the first and third term. If we choose to balance the first and last term, we get that $\delta \sim \varepsilon^{-1/3}$. Plugging this in we would get

$$y^3 - \varepsilon^{-1/3} y + 1 = 0$$

The first and third terms are now both order 1 but the middle term is huge as $\varepsilon \to 0$ which is impossible since you now have an unbalanced term. Therefore the correct terms to balance were the first and second. We then get $\delta \sim \frac{1}{\sqrt{\varepsilon}}$ and so

$$\frac{1}{\sqrt{\varepsilon}} y^3 - \frac{1}{\sqrt{\varepsilon}} y + 1 = 0 \to y^3 - y + \sqrt{\varepsilon} = 0.$$

We can now iterate/expand to find the roots we couldn't find earlier. Additionally, it is worth noting since the non-homogeneity is $\sqrt{\varepsilon}$, we should have our expansion be

$$y = y + 0 + \sqrt{\varepsilon} y_1 + (\sqrt{\varepsilon})^2 y_2 + \dots$$

3 Transcendental Equations

3.1 Expansion

Let us take the equation

$$x e^{-x} = \varepsilon$$

as an example. By simply plotting it, we see that there exists a root close to $x = 0$ and a large root as $x \to \infty$. One way to do it is to take the two term expansion for e^{-x} and then iterate on that. Doing so we get

$$\begin{aligned} x_{n+1} &= \varepsilon + x_n^2 \\ x_1 &= \varepsilon \\ x_2 &= \varepsilon + \varepsilon^2 \end{aligned}$$

To find the other root, we take the natural log of both sides and we get

$$\ln x - x = \ln \varepsilon$$

we then make the guess

$$x = \delta_0(\varepsilon) + \delta_1(\varepsilon) + \dots$$

where each of the basis functions are to be solved for on their own. We therefore arrive at

$$\ln(\delta_0) - \delta_0 = \ln \varepsilon$$

We want to balance the second and third terms because if we did the first and third, we would get that $\delta_0 = \varepsilon$ and we are trying to find the larger root. We then find that $\delta_0 = -\ln \varepsilon = \ln \frac{1}{\varepsilon}$. We now plug this δ_0 back into our ansatz and solve for δ_1

$$\ln\left(\delta_0\left(1 + \frac{\delta_1}{\delta_0}\right)\right) - \delta_0 - \delta_1 = \ln \varepsilon \ln \delta_0 + \ln\left(1 + \frac{\delta_0}{\delta_1}\right) - \delta_0 - \delta_1 = \ln \varepsilon \ln \delta_0 + \ln\left(1 + \frac{\delta_0}{\delta_1}\right) - \delta_1 = 0$$

We now want to balance the first and third term since first, it contains δ_1 which we are solving for, and secondly, δ_0 dominates over δ_1 the second term is close to zero. We then find that

$$\delta_1 = \ln\left(\ln \frac{1}{\varepsilon}\right)$$

It is also worth noting that previously we didn't care about signs or constants. That's because we were simply content with finding the same order of magnitude approximations. However, we are now looking for the *functions* for our expansion and so now we do care about these things.

3.2 Implicit Equations

Let us look at the equation

$$x^2 + xy - y^3 = 0$$

Say we want to know the asymptotic behavior of $y(x)$ as $x \to \infty$. We have three total ways to balance the terms in this equation:

1. $x^2 \leftrightarrow xy \Leftrightarrow x^2 + xy = 0 \to y = -x$
2. $xy \leftrightarrow y^3 \Leftrightarrow xy - y^3 = 0 \to y = \sqrt{x}$
3. $x^2 \leftrightarrow y^3 \Leftrightarrow x^2 - y^3 = 0 \to y = x^{2/3}$

If we substitute the each of the results we can check the balances to see which one is correct.

1. $x^2 \;\; xy \sim -x^2 \;\; y^3 \sim -x^3$ We see that as $x \to \infty$, the last term isn't balanced.
2. $x^2 \;\; xy \sim x^{3/2} \;\; y^3 \sim x^{3/2}$ This is the wrong choice as well since the x^2 term dominates even though we wanted the other two terms to dominate
3. $x^2 \;\; xy \sim x^{5/3} \;\; y^3 \sim x^2$ We see this is the correct balance because the balanced terms are the dominant terms in the equation.

We now guess that

$$y(x) = x^{2/3} + g(x)$$

where $g(x)$ is some correction term we seek to find. We want $g(x)$ to be less than the dominant term. Or in other words,

$$\lim_{x \to \infty} \frac{g}{x^{2/3}} \to 0$$

We now plug this into our equation and we get

$$x^2 + x^{5/3} + xg = x^2 + 3x^{2/3}g + 3x^{4/3}g + g^3$$

When we balance $x^{5/3} = 3x^{4/3}g$ we get that $g = \frac{1}{3}x^{1/3}$ which is the only correct balance (one can see for others by plugging in their relation and seeing if it matches what we desire). We now have that

$$x^{5/3} + x^{4/3} = 3x^{4/3} + 3x^{5/3} + x$$

and we see that our highest order terms are balanced which shows that are balance was correct. As such,

$$y(x) = x^{2/3} + \frac{1}{3}x^{1/3} + \dots$$

3.3 Approximating Integrals

Say we have some function $f(x, \varepsilon)$ which can be represented

$$f(x, \varepsilon) = \sum_n a_n(x)\varphi_n(\varepsilon)$$

We can integrate this function (however we can't differentiate because differentiation often causes the magnitude to increase with every derivative) and approximate the value of integrals. Let us take a look at the Stieltjes function (? can't find the name online but this is what Gemini gave)

$$I(\varepsilon) = \int_0^\infty \frac{e^{-t}}{1 + \varepsilon t} dt$$

We can define

$$I_n = \int_0^\infty \frac{e^{-t}}{(1 + \varepsilon t)^n} dt$$

If we differentiate by parts, we can derive the recurrence relation as follows

$$I_n = -\frac{e^{-t}}{(1+\varepsilon t)^n}\bigg|_0^\infty - n\varepsilon \int_0^\infty \frac{e^{-t}}{(1+\varepsilon t)^{n+1}} dt = 1 - n\varepsilon I_{n+1}$$

We find that $I_1 = 1 - \varepsilon I_2$. We also see that $I_2 = 1 - 2\varepsilon I_3$ and we can sub this back in to get that $I_1 = 1 - \varepsilon + 2\varepsilon^2 I_3$. We prefer this method to generate the integral approximation since we are adding smaller and smaller corrections with every approximation. We can ultimately arrive at

$$I(\varepsilon) = \sum_{n=0}^{N} (-1)\varepsilon^n n! + \text{error}(N)$$

3.4 Asymptotic Expansion

Let us say there exists a function such that

$$f(\varepsilon) = \sum_{n=1}^{N} a_n \varphi_n(\varepsilon) + \text{error}$$

where the error is $o(\varphi_N)$ for some basis of function space $\varphi_1, \varphi_2, \ldots, \varphi_N, \varphi_{N+1}, \ldots$. The interesting thing is that the series need not be convergent. Let us take a look at the error function

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

For small $t \ll 1$, we can Taylor expand such that

$$e^{-t^2} = 1 - t^2 + \ldots = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{n!}$$

We can integrate this Taylor expansion because integration is safer than differentiation. For example, if we take

$$f(\varepsilon) = \varepsilon \sin \frac{1}{\varepsilon} = \mathcal{O}(\varepsilon)$$

as $\varepsilon \to 0$ but if we differentiate, we end up getting a $\frac{1}{\varepsilon}$ which blows up. As such, we have that

$$\begin{aligned}
\text{erf}(x) &= \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^x t^{2n} dt \\
&= \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{x^{2n+1}}{2n+1} \\
&= \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3} + \ldots \right)
\end{aligned}$$

We now know the behavior of x close to 0, but what about as $x \to \infty$. Let us rewrite the error function in terms of the complementary error function

$$\text{erf}(x) = 1 - \text{erfc}(x) = 1 - \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

and then we integrate by parts to get

$$\text{erf}(x) = 1 - \frac{2}{\sqrt{\pi}} \left(-\frac{1}{2t} e^{-t^2} \bigg|_x^\infty - \frac{1}{2} \int_x^\infty \frac{e^{-t^2}}{t^2} dt \right)$$

which if we integrate by parts multiple times, we get the series

$$\text{erf}(x) = 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} \sum_{n=0}^{\infty} (-1)^n \frac{(2n-1)!!}{(2x^2)^n}$$

Sometimes, the series diverges and therefore adding more terms leads to a diverging from the true value. We can have the inverse of the radius of convergence to be

$$\frac{1}{R} = \lim_{n\to 0} \frac{a_{n+1}}{a_n} = \lim_{n\to 0} \frac{(2n+1)!!}{(2n-1)!!} = \lim_{n\to 0}(2n+1)$$

We now asks what is the optimal N that ensures convergence for the approximation $\sum_{n=0}^{N}$. This happens when $a_{N+1}\tilde{>}a_N$ and this is when we should stop. For a given x, we want to solve

$$(2x^2)^{-(N+1)}a_{N+1}\tilde{>}(2x^2)^{-N}a_N$$

which implies

$$\begin{aligned}\frac{a_{N+1}}{a_N}\frac{(2x^2)^N}{(2x^2)^{N+1}} &\geq 1\\ \frac{2n+1}{2x^2} &\geq 1\end{aligned}$$

where $N_{optimal}$ is the solution to the above inequality. We can get that

$$N_{opt} \sim \frac{1}{2} + x^2$$

For the transcendental equation

$$\tan x = x$$

we can see that the line $y = x$ contacts the tangent function right before the tangent blows up. As such, we can guess

$$x = A_n - \delta_n \qquad A_n = \frac{\pi}{2}(2n+1)$$

We can plug this into the tangent and express it as

$$\begin{aligned}\tan x &= \frac{\sin(A_n - \delta_n)}{\cos(A_n - \delta_n)}\\ &= \frac{\sin A_n \cos\delta_n - \cos A_n \sin\delta_n}{\cos A_n \cos\delta_n + \sin A_n \sin\delta_n}\\ &= \frac{\cos\delta_n}{\sin\delta_n}\end{aligned}$$

Taking the Taylor series of sines and cosines we have that

$$\tan x = \frac{1 - \frac{1}{2}\delta^2 + \mathcal{O}(\delta^4)}{\delta - \frac{1}{6}\delta^3 + \mathcal{O}(\delta^5)} = A - \delta$$

which if we pull a δ out of the denominator and and then cancel it

$$\begin{aligned}1 - \frac{1}{3}\delta^2 &= A\delta - \delta^2\\ A\delta &= 1 + \frac{2}{3}\delta^2\\ \delta &= \frac{1}{A}\left(1 + \frac{2}{3}\delta^2 + \dots\right)\end{aligned}$$

and if we use the leading order approximation $\delta \sim \frac{1}{A}$ which we then plug in

$$\delta = \frac{1}{A}\left(1 + \frac{2}{3}\frac{1}{A^2} + \mathcal{O}\left(\frac{1}{A^3}\right)\right)$$

4 Ordinary Differential Equations

4.1 Series Solutions

Consider the equation of the form

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

A natural question to ask are what are the solutions $y_1(x)$ and $y_2(x)$ around some point $x = x_0$. To do this, we need to consider three types of points:

1. Regular points like in $y'' + p(x)y' + q(x)y = 0$ where $p(x)$ and $q(x)$ are analytic at the point $x = x_0$
2. Regular singular points which have a "replaceable" singularity. This would be like

   $$(x - x_0)^2 R(x)y'' + (x - x_0)P(x)y' + Q(x)y = 0$$

   where the following limits are finite

   $$\lim_{x \to x_0} \frac{P(x)}{x - x_0} \qquad \lim_{x \to x_0} \frac{Q(x)}{(x - x_0)^2}$$

3. Irregular singular which often do not possess convergent series solutions.

Let us consider the Airy Equation

$$y'' - xy = 0 \qquad x_0 = 0$$

Going with the standard procedure, we guess $y(x) = \sum_{n=0}^{\infty} A_n x^n$. Differentiating twice and subbing it into the equation, we have

$$\sum_{n=2}^{\infty} A_n x^{n-2} n(n-1) - \sum_{n=0}^{\infty} A_n x^{n+1} = 0$$

We now want to shift the indices such that they match the same powers of x. We use the change of indices $n + 1 = m - 2 \to m = n + 3$ so we now have

$$\sum_{n=2}^{\infty} A_n x^{n-2} n(n+1) - \sum_{m=3}^{\infty} A_{m-3} x^{m-2} = 2A_2 + \sum_{n=3}^{\infty} (A_n n(n-1) - A_{n-3}) x^{n-3} = 0$$

Which then gets the recurrence relation

$$A_n = \frac{A_{n-3}}{n(n+1)}$$

We can then write the solution as

$$y = A_0 \left(1 + \frac{1}{6}x^3 + \dots\right) + A_1 \left(x + \frac{1}{12}x^4 + \dots\right)$$

where can think of each term in parenthesis as the "$y_1(x)$" and "$y_2(x)$." Let us take a look at an equation with a regular singular point

$$x^2 y'' - (x^2 + 2)y = 0 \qquad x_0 = 0$$

In this case, we use the Frobenius Method where we guess a solution of the form $y = x^s \sum_{n=0}^{\infty} A_n x^n$. Differentiating and substituting we get

$$\sum_{n=0}^{\infty} A_n (n+s)(n+s-1) x^{n+s} - \sum_{n=0}^{\infty} A_n x^{n+s+2} - 2\sum_{n=0}^{\infty} A_n x^{n+s}$$

We now want to make the change of indices using $n + 2 = m \to n = m - 2$ and thus our equation becomes

$$\sum_{n=0}^{\infty} A_n (n+s)(n+s-1) x^{n+s} - \sum_{n=2}^{\infty} A_{n-2} x^{n+s} - 2\sum_{n=0}^{\infty} A_n x^{n+s}$$

and by iterating the first values of n for the first and third series we get

$$(s(s-1)-2)A_0x^s + ((s+1)-2)A_1x^{s+1} + \sum_{n=2}^{\infty}(A_n((n+s)(n+s-1)-2) - A_{n-2})x^{n+s} = 0$$

The recurrence relation is therefore

$$A_n = \frac{A_{n-2}}{(n+s)(n+s-1)-2}$$

and we also get the indicial equation

$$s(s-1)-2 = 0 \to s = 2, -1$$

We therefore have our two solutions to be of the form

$$y_1 = x^2\sum_{n=2}^{\infty} A_n x^n$$
$$y_2 = \frac{1}{x}\sum_{n=2}^{\infty} A_n x^n$$

If we once again consider the Airy Equation, we can see that $x \to \infty$ is an essential singularity. To see this, we use the change of variables $t = \frac{1}{x}$ such that $y(x) \to y(t)$. Using the chain rule, we see that

$$\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = -t^2\frac{dy}{dt}$$

and

$$\frac{d}{dx}\left(-t^2\frac{dy}{dt}\right) = \frac{d}{dt}\left(-t^2\frac{dy}{dt}\right)(-t^2) = t^4y'' + 2t^3y'$$

Substituting this into the Airy Equation, we have that

$$y'' - xy = t^4y'' + 2t^3y' - \frac{1}{t}y = 0$$
$$\Rightarrow t^5y'' + 2t^4y' - y = 0$$

and we can see clearly that $x \to \infty$ is an essential singularity.

4.2 Asymptotics for ODEs

Let us go back to the original Airy's equation. Let us make the ansatz that our solution for large x is $y(x) = e^{S(x)}$ where $S(x) = S_1(x) + S_0(x) + \ldots$. Taking derivatives we have that $y'(x) = S'e^S$ and $y'' = S''e^S + (S')^2e^S$. Plugging this into Airy's Equation, we have

$$\begin{aligned}[S'' + (S')^2]e^S &= xe^S \\ \Rightarrow S'' + (S')^2 &= x\end{aligned} \tag{1}$$

We want large x and so we need to select a term to balance with it. If we balance the first and third term we get that $S'' \sim x \to S \sim x^3$ which if we substitute this balance into the second order differential equation, we would have that the first term $\sim x$, the second term $\sim x^4$, and the third would be $\sim x$. Since the second term, which is the term we don't want to balance, is larger than the two balanced terms, this is the incorrect choice for balancing. The correct balance is between the second and third terms which yields $S' = \sqrt{x} \to S = \frac{2}{3}x^{3/2}$ and $S'' = \frac{1}{2\sqrt{x}}$. Therefore we get the balance to be $\frac{1}{2\sqrt{x}}, x, x$ which we can see that the second and third terms are the highest order terms. We therefore have that the leading order behavior for large x is

$$y = e^{2x^{3/2}/3}$$

Now we plug in the next order approximation. We therefore have

$$\left(\frac{1}{2\sqrt{x}} + S_1''\right) + \left(\sqrt{x} + S_1'\right)^2 = x$$
$$\frac{1}{2\sqrt{x}} + S_1'' + x + (S_1')^2 + 2\sqrt{x}S_1' = x$$
$$S'' + (S_1')^2 + s\sqrt{x}S_1' + \frac{1}{2\sqrt{x}} = 0$$

If we take a balance like $S_1' \sim -2\sqrt{x} \to S_1 \sim -\frac{4}{3}x^{3/2}$, this would be incorrect because we can't have the correction term be the same order as the leading order term. In other words, we require that this correction term needs to be $o(x^{3/2})$. The correct balance is between the third and fourth terms which yields $-2\sqrt{x}S_1' = \frac{1}{2\sqrt{x}} \to S_1' = -\frac{1}{4x} \to S_1 = -\frac{1}{4}\ln x$. Our two term expansion for large x yields

$$y(x) = \exp\left(\frac{2}{3}x^{3/2} - \frac{1}{4}\ln x\right)$$

Let us recall the rocket/projectile problem in the motivating example. We can describe the height $y(t)$ of a projectile given some initial velocity v_0 with the following differential equation

$$\frac{d^2y}{dt^2} = -\frac{1}{(1+\varepsilon y)^2} \qquad 0 < \varepsilon = \frac{v_0^2}{gR_e} \ll 1$$

We guess our y in the form $y = y_0 + \varepsilon^\alpha y_1 + \ldots$. Using $\varepsilon = 0$, we find the leading order term is described by

$$\frac{d^2y_0}{dt^2} = -1 \qquad y_0(0) = 0 \qquad y_0'(0) = 1$$

where we have chosen the initial velocity to be 1 as it is the "characteristic" velocity and therefore is the non-dimensionalized form. This yields the leading term behavior to be $y_0(t) = t - \frac{1}{2}t^2$. Moving onto the next order, let us use the binomial expansion

$$\frac{1}{(1+z)^2} \approx 1 - 2x$$

to get that

$$\frac{d^2y_0}{dt^2} + \varepsilon\frac{d^2y_1}{dt^2} = -1 + 2\varepsilon(y_0 + \varepsilon y_1 + \ldots)$$

Grouping up the terms based off their orders with respect to ε, we have that

$$\mathcal{O}(1): \qquad \frac{d^2y_0}{dt^2} = -1$$
$$\mathcal{O}(\varepsilon): \qquad \frac{d^2y_1}{dt^2} = 2y_0$$

We also have the initial conditions to be

$$y(0) = 0 = y_0(0) + \varepsilon y_1(0) + \ldots$$
$$y'(0) = 1 = y_0'(0) + \varepsilon y_1'(0) + \ldots$$

We notice that the leading term takes out the inhomogeneity and the later correction terms become homogenous initial conditions. This leads to the first correction term becoming

$$y_1(t) = \frac{1}{3}t^3 - \frac{1}{12}t^4$$

and so our combined solution is

$$y(t) = y_0(t) + \varepsilon y_1(t) = t - \frac{1}{2}t^2 + \varepsilon\left(\frac{1}{3}t^3 - \frac{1}{12}t^4\right)$$

5 Boundary Layers

5.1 Singular Perturbation

A singular perturbation occurs when we have some sort of equation like

$$\varepsilon y'' + y' + y = 0 \qquad y(0) = 0 \qquad y(1) = 1$$

For the $\varepsilon = 0$, we get that $y_0' + y_0 = 0$ which implies $y_0(x) = Ce^{-x}$ but we now have 2 boundary conditions. We can't match both boundary conditions so we introduce a rescaling such that $\eta = \frac{x}{\varepsilon^\alpha}$. We now shift our function $y(x)$ to $Y(\eta)$ and $\frac{dy}{dx} = \frac{dY}{d\eta}\frac{d\eta}{dx}$ where $\frac{d\eta}{dx} = \varepsilon^{-\alpha}$. We therefore have that our singular perturbed equation becomes

$$\varepsilon^{-2\alpha+1}Y'' + \varepsilon^{-\alpha}Y' + Y = 0 \tag{2}$$

We now see once small second derivative is now very large and we want to balance it. The correct balance is with the second term which implies $\varepsilon^{-2\alpha+1} = \varepsilon^{-\alpha} \to \alpha = 1$. This leads to

$$\begin{aligned}\varepsilon^{-1}Y'' + \varepsilon^{-1}Y' + Y &= 0\\ Y'' + Y' + \varepsilon Y &= 0\end{aligned}$$

For this, we see the leading order behavior to be

$$Y_0(\eta) = A_0(1 - e^{-\eta})$$

where we have the initial conditions $x = 0, y(0) = 0$ which implies $Y_0(0) = 0$. This equation above is known as the *inner* solution since it captures the behavior near the boundary layer which isn't captured by the *outer* solution which is

$$y_0(x) = e^{1-x}$$

These solutions are *non-uniform* approximations because for some values, they deviate significantly from the true solution value. What we are looking for is a *uniform* approximation. We do this by "stitching" the inner and outer solution together by matching asympotitcs. In this case we take $x \to 0$ and $\eta \to \infty$ since the boundary layer is on the left.

$$\lim_{x\to 0} y(x) = \lim_{\eta\to\infty} Y_0(\eta)$$

Substituting our inner and outer solutions, we get $A_0 = e$. We then write our uniform approximation as

$$y(x) = \text{inner} + \text{outer} - \text{matching term}$$

And we subtract the matching constant to prevent double counting. So our completed solution is

$$\begin{aligned}y_0(x) &= e(1 - e^{-x/\varepsilon}) + e^{1-x} - e\\ &= e^{1-x} - e^{1-x/\varepsilon}\end{aligned}$$

Let us take a step back and consider the general case of

$$\varepsilon y'' + p(x)y' + q(x)y = 0 \qquad y(a) = A \qquad y(b) = B \qquad a < x < b$$

Our outer solution is of the form

$$y \sim \exp\left[-\int \frac{q(x)}{p(x)}dx\right]$$

If we suppose the boundary layer is on the left, we make the coordinate change $\eta = \frac{x-a}{\varepsilon}$. Our equation then becomes

$$\begin{aligned}Y'' + p(a + \varepsilon\eta)Y' + \varepsilon q(a + \varepsilon\eta)Y &= 0\\ Y_0'' + p(a)Y_0' &= 0\\ Y_0 &\sim \exp[-p(a)\eta]\end{aligned}$$

For this to generate our desired uniform solution, we need Y_0 to stay finite as $\eta \to \infty$ and therefore $p(a) > 0$ if the boundary layer is on the left. In a similar fashion, we can see that we require $p(b) < 0$ if the boundary layer is on the right. Let us take a look at the boundary value problem

$$\varepsilon y'' - (1 + x)y' + 3y = 0 \qquad y(0) = y(1) = 1$$

We can see that $p(x) = -(1 + x)$ and since $p(0) < 0$, we cannot have the boundary layer be on the left and therefore it is at the right. We can either make the change of variables $\eta = \frac{1-x}{\varepsilon}$ with $\eta \to \infty$ or $\eta = \frac{x-1}{\varepsilon}$ with $\eta \to -\infty$. The outer solution is

$$\begin{aligned}-(1 + x)y_0' + 3y_0 &= 0\\ y_0 &= (1 + x)^3\end{aligned}$$

If we're using the $\frac{\eta-1}{\varepsilon}$ scheme, we have $x = 1 + \varepsilon\eta$. Therefore for the inner solution, we have

$$\begin{aligned}Y'' - (2 + \varepsilon\eta)Y' + 3\varepsilon Y &= 0\\ Y_0'' - 2Y_0' &= 0\\ Y_0 &= A_0(e^{2\eta} - 1) + 1\end{aligned}$$

From this, we see that

$$\begin{aligned}\lim_{x\to 1}(1+x)^3 &= \lim_{\eta\to-\infty} A_0(e^{2\eta}-1)+1\\ 8 &= 1-A_0\\ A_0 &= -7\end{aligned}$$

And therefore our composite solution is

$$y(x) = (x+1)^3 - 7(e^{\frac{x-1}{\varepsilon}}-1)$$

Let us consider an example where $\alpha \neq 1$.

$$\varepsilon y'' + x^{1/2}y' - y = 0 \qquad y(0)=0 \qquad y(1)=e^2$$

We start by noting that the boundary layer is on the left. This leads the outer solution being

$$y_0 = e^{2\sqrt{x}}$$

We apply the change of variables $\eta = \frac{x}{\varepsilon^\alpha}$ and this gets us

$$\varepsilon^{-2\alpha+1}Y'' + \varepsilon^{-\alpha/2}\sqrt{\eta}Y' - Y = 0$$

Balancing the first two terms, we get that $1-2\alpha = -\frac{\alpha}{2} \to \alpha = \frac{2}{3}$. We then get

$$Y_0'' + Y_0'\sqrt{\eta} = 0$$

Therefore integrating twice gives us

$$Y_0 = C\int_0^\eta e^{-2\eta'^{3/2}/3}d\eta'$$

5.2 Van Dyke Method

If we want to consider a two term expansion, we have to be a little more careful. Let us consider the equation

$$\varepsilon y'' + y' = x \qquad y(1)=1 \qquad y(0)=0$$

The boundary layer is on the left and skipping some steps, we get that

$$y_{\text{out}}(x) = \frac{1}{2}(1+x^2) + \varepsilon(1-x) + o(\varepsilon)$$

and the inner solution to be

$$Y_{\text{inner}}(\eta) = C_0(1-e^{-\eta}) + \varepsilon C_1(1-e^{-\eta}) + o(\varepsilon)$$

with our change of variables $\eta = \frac{x}{\varepsilon}$ (since $\alpha = 1$). We rewrite one of the solutions in terms of the other variable. In this case, we rewrite the outer solution in terms of the inner variable to get

$$\begin{aligned}y_{\text{out}}(\varepsilon\eta) &= \frac{1}{2}(1+\varepsilon^2\eta^2) + \varepsilon(1-\varepsilon\eta)\\ &= \frac{1}{2} + \varepsilon + \mathcal{O}(\varepsilon^2)\end{aligned}$$

and now we let $\varepsilon \to 0$ which we then get that $C_0 = \frac{1}{2}$ and $C_1 = 1$. We now find the uniform solution as

$$y(x) = y_{\text{out}}(x) + Y_{\text{in}}\left(\frac{x}{\varepsilon}\right) - \text{matching}$$

and therefore

$$y(x) = \frac{1+x^2}{2} + \varepsilon(1-x) - \left(\frac{1}{2}+\varepsilon\right)e^{-x/\varepsilon}$$

5.3 Exotic Boundary Layers

If the coefficient of the first derivative term changes sign over the interval, then there exists a turning point at some value x_0. We let this coefficient $p(x)$ be equal to some $\alpha(x - x_0)$ where α is some constant. If we just take $x_0 = 0$ on some interval $[-1, 1]$, then $p \sim \alpha x$. If $\alpha > 0$ and $x < 0$, then $p(x) < 0$ implies there's a boundary layer on the right. If $x > 0$, then $p(x) > 0$ which implies there's a boundary layer on the left and so hence there is an internal boundary layer at x_0. When $\alpha < 0$, then we have that $x < 0 \to p(x) > 0$ which implies a boundary layer on the left. For $x > 0$, $p(x) < 0$ which implies a boundary on the right. Hence, there are two boundary layers. Dealing with these boundary layers is same as the procedure above but we have to match the two outer solutions with the inner for an internal and the one outer with both inner solutions for double boundary layers.

There is another case where one might have nested boundary layers. This occurs when there exist multiple dominant balances. Another sign of a nested boundary layer is when you cannot match the inner and outer solutions. One has to be careful that the inner boundary layer is the thinnest one. As such, we must create some intermediate solution with say some sort of new variable ξ and function $Z(\xi)$. Then the matching should be that $\lim_{\eta\to\infty} Y = \lim_{\xi\to 0} Z$ and $\lim_{\xi\to\infty} Z = \lim_{x\to 0} y$ as an example.

6 Secular Terms

If we consider the Duffing equation given by

$$\ddot{x} + x + \varepsilon x^3 = 0 \qquad x(0) = 1 \qquad x'(0) = 0$$

we cannot just do a regular perturbation since the leading order approximation yields

$$\mathcal{O}(1) : \ddot{x}_0 + x_0 = 0 \to x_0 = \cos t$$

and the first order correction gives rise to a secular term

$$\begin{aligned}\mathcal{O}(\varepsilon) : \ddot{x}_1 + x_1 = -x_0^3 \to x_1 &= \frac{1}{32}\left(\cos 3t - \cos t - 12t \sin t\right) \\ \Rightarrow x(t) &= \cos t + \frac{\varepsilon}{32}\left(\cos 3t - \cos t - 12t \sin t\right)\end{aligned}$$

This approximation starts breaking down around when $t \approx \frac{1}{\varepsilon}$ since that is when the secular term becomes non-negligible. To resolve this, we want to expand the period in orders of ε.

6.1 Method of Strained Coordinates

We start with the ansatz that $\omega = \omega_0 + \varepsilon\omega_1 + \ldots$ and $\tau = \omega t$. Then we have $x(t) \to X(\tau)$. Therefore $\frac{dX}{dt} = \frac{dX}{d\tau}\frac{d\tau}{dt} = \omega\frac{dX}{d\tau}$. The Duffing Equation now becomes

$$\omega^2 \frac{d^2X}{d\tau^2} + X + \varepsilon X^3 = 0 \qquad X(0) = 1 \qquad \omega\frac{dX}{d\tau}(0) = 0$$

The leading order equation is the same as before and $x_0 = \cos t$. The first correction, after expanding ω and X, we get that

$$\ddot{X}_1 + 2\omega\ddot{X}_0 + X_1 + X_0^3 = 0$$

and hence we are left with

$$\ddot{X}_1 + X_1 = -X_0^3 - 2\omega_1\ddot{X}_0$$

Expanding the right hand side and using the identity $\cos^3 t = \frac{1}{4}\cos 3t + \frac{3}{4}\cos t$, we get

$$-X_0^3 - 2\omega_1\ddot{X}_0 = \left(2\omega_1 - \frac{3}{4}\right)\cos\tau - \frac{1}{4}\cos 3\tau$$

and we want $2\omega_1 - \frac{3}{4} = 0$ as the natural frequency of the oscillator on the left hand side is $\omega = 1$. Therefore we want to eliminate any term with the same frequency as the natural frequency of the oscillator to prevent resonance. Therefore $\omega = \frac{3}{8}$. And therefore the first correction is given by

$$x_1(\tau) = \frac{1}{32}(\cos 3\tau - \cos tau)$$

and the leading order plus first oder approximation is

$$x(t) = \cos\left(\left(1+\frac{3}{8}\varepsilon\right)t\right) + \frac{\varepsilon}{32}\left[\cos\left(3\left(1+\frac{3}{8}\varepsilon\right)t\right) + \cos\left(\left(1+\frac{3}{8}\varepsilon\right)t\right)\right]$$